

Unit - 1

Error

Difference between the actual value (X) and approximate value (X')

Accuracy

Accuracy is how close a measured value is to the actual value.

Precision

Precision is how close the measured values are to each other.

Eg: If you are playing football, and you always hit the right goal post instead of scoring, then you are ~~a~~ not accurate but precise.

Accurate = Correct
Precise = Repeating

Absolute Error

$$E_a = X - X'$$

X = Actual value

X' = Approximate Value

Relative Error

$$E_r = \left| \frac{X - X'}{X} \right|$$

%age Error

$$E_p = \left| \frac{X - X'}{X} \right| \times 100\%$$

Unit-2 Theory

Algebraic Equation

An equation $f(x) = 0$ is called an algebraic equation if $f(x)$ is an algebraic polynomial.

eg: $x^3 - 4x - 9 = 0$.

Transcendental Equation

If $f(x)$ contains functions such as trigonometric, logarithmic, exponential etc, $f(x)$ is called a transcendental equation.

eg: $2x^3 - \log(x+3) \tan x + e^x = 0$.

Rate of convergence

1. Bisection-Method

Rate of convergence is linear.
(slowest)

2. Regula-falsi

Rate of convergence is linear.
(faster than Bisection)

3. Secant Method

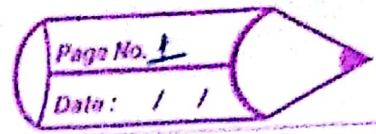
Rate of convergence = 1.6
(if it converges once, which is faster than false position)

4. N-R Method

Rate of convergence = quadratic and convergence is of order 2.

Unit-2

Roots of Equation



① Bisection-Method

$f(x) = 0$ be the given eqⁿ

$$f(x_1) < 0 \quad \& \quad f(x_0) > 0$$

$\Rightarrow$ Root lies in $[x_1, x_0]$

{ 1st -ve to the value corresponding x is the interval for roots? }

$$x_m = \frac{a+b}{2}$$

If $f(x_0) = 0 \Rightarrow x_0$ is root & stop.

If $f(x_0) < 0 \Rightarrow$ Root lies $[x, x_0]$ [Left change]

If $f(x_0) > 0 \Rightarrow$ Root lies $[x_1, x]$ [Right change]

Note:-

Log Base Change Rule

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Q] $x^3 - 5x + 3 = 0$ upto 4 decimal places.

Step 1: $f(0) = 0 - 0 + 3 = 3 > 0$
 $f(1) = 1 - 5 + 3 = -1 < 0$

$\therefore$ interval is $[1, 0]$ { -ve to +ve }

Step-1 $x_1 = \frac{1+0}{2} = 0.5$

$$f(x_1) = (0.5)^3 - 5(0.5) + 3$$
$$f(0.5) = 0.62500 > 0$$

Step-2 Root lies b/w 0.5 and 1 $[1, 0.5]$

$$x_2 = \frac{1+0.5}{2} = 0.75000$$

$$f(x_2) = (0.75)^3 - 5(0.75) + 3$$
$$f(0.75) = -0.32813 < 0$$

Step-3 Root lies b/w 0.75 and 0.5 $[0.75, 0.5]$

$$x_3 = \frac{0.75+0.5}{2} = 0.62500$$

$$f(x_3) = (0.625)^3 - 5(0.625) + 3$$
$$f(0.625) = 0.11914 > 0$$

Step-4 Root lies b/w 0.75 and 0.625 $[0.75, 0.625]$

$$x_4 = \frac{0.75+0.625}{2} = 0.68750$$

$$f(x_4) = (0.68750)^3 - 5(0.68750) + 3$$
$$f(0.68750) = -0.11255 < 0$$

Sta-5 Root lies in 0.625 & 0.6875 [0.6875, 0.625]

$$x_5 = \frac{0.6875 + 0.625}{2} = 0.65625$$

$$f(x_5) = 0.00137 > 0$$

Sta-6 Root lies in 0.65625 & 0.6875 [0.6875, 0.65625]

$$x_6 = \frac{0.6875 + 0.65625}{2} = 0.67188$$

$$f(x_6) = -0.05608 < 0$$

Sta-7 Root lies in 0.65625 & 0.67188 [0.67188, 0.65625]

$$x_7 = \frac{0.67188 + 0.65625}{2} = 0.66407$$

$$f(x_7) = -0.02748 < 0$$

Sta-8 Root lies in 0.65625 & 0.66407 [0.66407, 0.65625]

$$x_8 = \frac{0.66407 + 0.65625}{2} = 0.66016$$

$$f(x_8) = -0.01309 < 0$$

Sta-9 Root lies in 0.65625 & 0.66016 [0.66016, 0.65625]

$$x_9 = \frac{0.66016 + 0.65625}{2} = 0.65821$$

$$f(x_9) = -0.00587 < 0$$

Sta-10 Root lies in 0.65625 & 0.65821 [0.65821, 0.65625]

$$x_{10} = \frac{0.65821 + 0.65625}{2} = 0.65723$$

$$f(x_{10}) = -0.00226 < 0$$

Sta-11 Root lies in 0.65625 & 0.65723 [0.65723, 0.65625]

$$x_{11} = \frac{0.65625 + 0.65723}{2} = 0.65674$$

$$f(x_{11}) = -0.00044 < 0$$

Sta-12 Root lies in 0.65625 & 0.65674 [0.65674, 0.65625]

$$x_{12} = \frac{0.65674 + 0.65625}{2} = 0.65650$$

$$f(x_{12}) = 0.00046 > 0$$

Sta-13 Root lies in 0.65650 & 0.65674 [0.65674, 0.65650]

$$x_{13} = \frac{0.65674 + 0.65650}{2} = 0.65662$$

$$f(x_{13}) = 0.00000$$

② Newton-Raphson Method

$f(x) = 0$ be the given eqⁿ

x_0 = approx root of eqⁿ

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} ; n = 0, 1, 2, 3, \dots$$

Step 1: Find interval I.

Step 2: Find $f'(x)$

Step 3: Insert into formula.

Take initial approximation x_0 as any one value from the interval values.
(Hint: Take the nearest approx. ans $\pm$)

- Find interval I the same way as for Bisection.

- Convert degree - radian in calculator when trigonometry sums given.

Q] $f(x) = x \sin x + \cos x$. near $x = \pi$ upto 4.

Here, Given $x_0 = \pi$

Stk 1: $n=0$

$$\begin{aligned} f'(x) &= x \cos x + \sin x - \sin x \\ f'(x) &= x \cos x \end{aligned}$$

$$x_{0+1} = x_0 - \frac{f(x_0)}{f'(x_0)} \quad \begin{aligned} \cos \pi &= -1 \\ \sin \pi &= 0 \end{aligned}$$

$$= \pi - \frac{[\pi \sin \pi + \cos \pi]}{\pi \cos \pi}$$

$$x_1 = \pi - \frac{(-1)}{-\pi} = \pi - \frac{1}{\pi}$$

$$x_1 = 2.82328$$

Stk 2 $n=1$

$$x_{1+1} = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 2.82328 - \frac{f(2.82328)}{f'(2.82328)}$$

$$= 2.82328 - \left(\frac{-0.06618}{-2.68145} \right)$$

$$x_2 = 2.79860$$

Stk 3 $n=2$

$$x_{2+1} = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = 2.79860 - \frac{f(2.79860)}{f'(2.79860)}$$

$$= 2.79860 - \left(\frac{-0.00056}{-2.63559} \right)$$

$$x_3 = 2.79839$$

Q1-4 $x_1 = 3$

$$x_{3+1} = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$= 2.79839 - \frac{f(2.79839)}{f'(2.79839)}$$

$$x_4 = 2.79839 - 0$$

$\therefore$ Root upto 4 decimal places is 2.7983.

Q] $f(x) = x^4 - x - 10$ upto 3 decimal places.

$f(0) = 0 - 0 - 10 = -10 < 0$	$f'(x) = 4x^3 - 1$
$f(1) = 1 - 1 - 10 = -10 < 0$	
$f(2) = 16 - 2 - 10 = 4 > 0$	

$$\therefore \text{interval } I = [1, 2]$$

Let $x_0 = 2$ (i.e. initial approximation)

Q1-1 $x = 0$

$$x_{0+1} = x_0 - \frac{f(x_0)}{f'(x_0)}$$

(P.T.O)

$$x_1 = 2 - \frac{f(2)}{f'(2)} = 2 - \frac{4}{3}$$

$$x_1 = 1.8710$$

St2-2 $n=1$

$$x_{1+1} = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 1.8710 - \frac{f(1.8710)}{f'(1.8710)}$$

$$x_2 = 1.8710 - \frac{0.3827}{25.1988}$$

$$x_2 = 1.8558$$

St2-3 $n=2$

$$x_{2+1} = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.8558 - \frac{f(1.8558)}{f'(1.8558)}$$

$$= 1.8558 - \frac{0.0056}{24.5655}$$

$$= 1.8556$$

∴ root upto 3 decimal places = 1.855

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③ Deductions by N-R Method

① $\frac{1}{N} \Rightarrow x_{n+1} = x_n [2 - Nx_n]$

② $\frac{1}{\sqrt{N}} \Rightarrow x_{n+1} = \frac{1}{2} \left[x_n + \frac{1}{Nx_n} \right]$

③ $\sqrt[k]{N} \Leftrightarrow (N)^{1/k} \Rightarrow x_{n+1} = \frac{1}{k} \left[(k-1)x_n + \frac{N}{(x_n)^{k-1}} \right]$

Q] $\sqrt{28}$ upto 3 decimal places.

$$x_{n+1} = \frac{1}{k} \left[(k-1)x_n + \frac{N}{(x_n)^{k-1}} \right]$$

here, $k = 2$ ($\because \sqrt{28} = (28)^{1/2}$) $\left[\frac{1}{2} = \frac{1}{k} \right]$
($\Rightarrow k=2$)

$$x_{n+1} = \frac{1}{2} \left[x_n + \frac{N}{x_n} \right]$$

Initial approximation $x_0 = 5$

[Always take near by whole no as initial approximation here $28 = 5^2$ thus $x_0 = 5$]

1st-1 $n=0$

$$x_{0+1} = \frac{1}{2} \left[x_0 + \frac{N}{x_0} \right]$$

$$= \frac{1}{2} \left[5 + \frac{28}{5} \right]$$

$$x_1 = 5.6000$$

x_1-2 $n=1$

$$x_{1+1} = \frac{1}{2} \left[x_1 + \frac{N}{x_1} \right]$$

$$= \frac{1}{2} \left[5.6 + \frac{28}{5.6} \right]$$

$$x_2 = 5.3000$$

(P.T.O)

Ex-3

$$n=2$$

$$x_{n+1} = \frac{1}{2} \left[x_n + \frac{N}{x_n} \right]$$

$$= \frac{1}{2} \left[5.3 + \frac{28}{5.3} \right]$$

$$x_3 = \cancel{5.44506} \quad 5.2915$$

Ex-4

$$n=3$$

$$x_{n+1} = \frac{1}{2} \left[x_n + \frac{N}{x_n} \right]$$

$$= \frac{1}{2} \left[5.2915 + \frac{28}{5.2915} \right]$$

$$x_4 = 5.2915$$

∴ Root is 5.291

Q] $\sqrt[3]{58}$ upto 4 decimal places.

$$x_{n+1} = \frac{1}{k} \left[(k-1)x_n + \frac{N}{(x_n)^{k-1}} \right]$$

here $k=3$ & $N=58$

Let $x_0 = 4$.

Ex-1 $n=0$

$$x_{0+1} = \frac{1}{3} \left[(3-1)x_0 + \frac{N}{(x_0)^{(3-1)}} \right]$$

$$= \frac{1}{3} \left[2 \times 4 + \frac{58}{(4)^2} \right]$$

$$x_1 = 3.8750$$

Ex-2 $n=1$

$$x_{1+1} = \frac{1}{3} \left[(2)(3.8750) + \frac{58}{(3.8750)^2} \right]$$

$$x_2 = 3.8709$$

Ex-3 $n=2$

$$x_{2+1} = \frac{1}{3} \left[(2)(3.8709) + \frac{58}{(3.8709)^2} \right]$$

$$x_3 = 3.8709$$

$\therefore$ Upto 4 decimal root is 3.8709

Q] $\frac{1}{\sqrt{14}}$ upto 4 decimal

$$N = 14.$$

$$x_{n+1} = \frac{1}{2} \left[x_n + \frac{1}{N \cdot x_n} \right]$$

$$\left\{ \begin{array}{l} \text{Let} \\ x_0 = \frac{1}{\sqrt{16}} \\ = \frac{1}{4} \end{array} \right.$$

Ex-1 $n=0$

$$\begin{aligned} x_{0+1} &= \frac{1}{2} \left[x_0 + \frac{1}{N x_0} \right] \\ &= \frac{1}{2} \left[\frac{1}{4} + \frac{1}{14 \left(\frac{1}{4} \right)} \right] \end{aligned}$$

$$x_1 = 0.2679$$

Ex-2 $n=1$

$$x_{1+1} = \frac{1}{2} \left[x_1 + \frac{1}{N x_1} \right]$$

$$x_2 = \frac{1}{2} \left[0.2679 + \frac{1}{14(0.2679)} \right]$$

$$x_2 = 0.2673$$

Ex-3 $n=2$

$$x_{2+1} = \frac{1}{2} \left[x_2 + \frac{1}{N x_2} \right]$$

$$x_3 = \frac{1}{2} \left[0.2673 + \frac{1}{14(0.2673)} \right]$$

$$x_3 = 0.2673$$

$\therefore$ Root = 0.2673 upto 4 decimal places.

Q] $\frac{1}{31}$ upto 4 decimal places.

$$N = 31$$

Iterative formula $x_{n+1} = x_n [2 - Nx_n]$

Let initial approximation $x_0 = \frac{1}{30}$

Step-1 $n=0$

$$x_{0+1} = x_0 [2 - (31)(x_0)]$$

$$x_1 = \frac{1}{30} \left[2 - \frac{31}{30} \right] \quad \therefore x_1 = 0.03222$$

Step-2 $n=1$

$$x_{1+1} = x_1 [2 - (31)(x_1)]$$

$$= (0.03222) [2 - (31)(0.03222)]$$

$$\therefore x_2 = 0.03226$$

Step-3 $n=2$

$$x_{2+1} = x_2 [2 - (31)(x_2)]$$

$$= (0.03226) [2 - (31)(0.03226)]$$

$$\therefore x_3 = 0.03226$$

Approx ans upto 4 decimal = 0.0322

④ Regula - Falsi - Method

Step 1: Find interval I as in Bisection Method.

Step 2: $I = [a, b]$

Step 3: Iterations

$$\text{Formula: } \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

Step 4: Get the root upto asked decimal places.

Here, also same as bisection ,
 $f(x) < 0 \Rightarrow [\text{left change}, \gamma]$
 $f(x) > 0 \Rightarrow [\gamma, \text{right change}]$

Q] $f(x) = x^3 - 4x + 1$ upto 3 decimal places.

$$f(0) = 0 - 0 + 1 = 1 > 0$$

$$f(1) = 1 - 4 + 1 = -2 < 0$$

$$\therefore \text{interval } I = [1, 0] = [a, b]$$

Step-1 Root lies in $[1, 0] = [a, b]$

$$x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

$$x_1 = \frac{(1)f(0) - 0 f(1)}{f(0) - f(1)}$$

$$x_1 = \frac{(1)(1) - 0(-2)}{1 - (-2)}$$

$$x_1 = \frac{1}{3} = 0.3333$$

$$f(x_1) = -0.2963 < 0$$

Step-2 Root lies in $[0.3333, 0] = [a, b]$

$$x_2 = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

$$= \frac{0.3333 - 0}{1 - (-0.2963)}$$

$$x_2 = 0.2571$$

$$f(x_2) = -0.0115 < 0$$

Str-3 Root lies in $[0.2571, 0] = [a, b]$

$$x_3 = \frac{0.2571 f(0) - 0 f(0.2571)}{f(0) - f(0.2571)}$$

$$= \frac{(0.2571)(1) - 0}{1 - (-0.0115)}$$

$$x_3 = \cancel{0.2604} 0.2542$$

Str-4 $f(x_3) = -0.000078 < 0$

Str-4 Root lies in $[0.2542, 0] = [a, b]$

$$x_4 = \frac{0.2542 f(0) - (0) f(0.2542)}{f(0) - f(0.2542)}$$

$$= \frac{0.2542}{1 - (-0.0003)}$$

$$= 0.2541$$

∴ Root upto 3 decimal places = 0.254.

⑤ Secant Method

step 1: Find interval I

step 2: $I = [x_n, x_{n+1}]$

step 3: Use iterative formula

Formula:

$$x_{n+1} = x_n - \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right] \times f(x_n)$$

$n \geq 1$

step 4: $f(x) < 0$ change left limit
 $f(x) > 0$ change right limit

[If initial values x_0 & x_1 are given then use them directly].

Q]

$x - 2 \sin x = 0$ upto 3 decimal places.
starting from $x_0 = 2$ & $x_1 = 1.9$
 $f(0) = 0 - 2 \sin(0) = 0$
 $f(1) =$

$$f(x_0) = f(2) = 2 - 2 \sin(2) = 0.1814$$

$$f(x_1) = f(1.9) = 1.9 - 2 \sin(1.9) = 0.0074.$$

By secant method,

$$x_{n+1} = x_n - \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right] \times f(x_n)$$

Step-1

$n=1$

Root lies in $[1.9, 2]$

$$x_{1+1} = x_1 - \left[\frac{x_1 - x_0}{f(x_1) - f(x_0)} \right] \times f(x_1)$$

$$x_2 = 1.9 - \left[\frac{1.9 - 2}{0.0074 - 0.1814} \right] \times 0.0074$$

$$x_2 = 1.8957$$

$$f(x_2) = 0.0004 > 0$$

Ex-2 $x = 2$

$$x_{2+1} = x_2 - \left[\frac{x_2 - x_1}{f(x_2) - f(x_1)} \right] \times f(x_2)$$

$$= 1.8957 - \left[\frac{1.8957 - 1.9}{0.0004 - 0.0074} \right] \times 0.0004$$

$$x_3 = 1.8955$$

Root upto 3 decimal places = 1.895

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⑥ Iteration Method [Successive Approximation Method]

Step 1: Find interval $I = [a, b]$

Step 2: Find $\phi(x)$
i.e. funcⁿ of x

Step 3: Fulfill condition that $|\phi'(x)| < 1$
for all search values in I .
[Hint: Check for $|\phi'(a)| < 1$ and $|\phi'(b)| < 1$]

Step 4: When (step-3) validates apply the iterative method

Formula:

$$x_{n+1} = \phi(x_n)$$

Step 5: Continue iterations till the required answer is obtained upto given/asked decimal places.

Q] $f(x) = x^3 + x^2 - 1$ upto 4 decimal places.

$$f(0) = 0 + 0 - 1 = -1 < 0$$

$$f(1) = 1 + 1 - 1 = 1 > 0$$

$$\text{interval } I = [0, 1]$$

$$x^3 + x^2 - 1 = 0$$

$$x^2 = 1 - x^3$$

$$\therefore x = (1 - x^3)^{1/2}$$

$$\therefore \cancel{x} = \cancel{(1 - x^3)^{1/2}}$$

$$\text{say } \phi(x) = (1 - x^3)^{1/2}$$

$$\phi'(x) = \frac{1}{2} (1 - x^3)^{\frac{1}{2} - 1} (-3x^2)$$

$$= -\frac{3}{2} (1 - x^3)^{-1/2} (x^2)$$

$$\phi'(0) = 0 \quad |\phi'(0)| < 1$$

$$\phi'(1) = -\frac{3}{2} (1 - 1)^{-1/2} (1) = \infty \quad |\phi'(1)| \neq 1$$

Hence, not valid.

$$x^3 + x^2 - 1 = 0$$

$$x^2(x^2 + 1) - 1 = 0$$

$$x^2(x^2 + 1) = 1$$

$$x^2 = \frac{1}{x^2 + 1}$$

$$x = \frac{1}{\sqrt{x^2 + 1}}$$

$$\phi(x) = (x^2 + 1)^{-1/2}$$

$$\phi'(x) = -\frac{1}{2} (x^2 + 1)^{-\frac{1}{2} - 1}$$

$$= -\frac{1}{2} (x^2 + 1)^{-3/2}$$

$$= -\frac{1}{2} \frac{1}{(x^2 + 1)^{3/2}}$$

$$\phi'(x) = -\frac{1}{2} \left(\frac{1}{(x^2 + 1)^{3/2}} \right)$$

$$\phi'(0) = -\frac{1}{2} \quad |\phi'(0)| < 1$$

$$\phi'(1) = -\frac{1}{2^{5/2}} \quad |\phi'(1)| < 1$$

Valid,

hence $\phi(x) = \frac{1}{\sqrt{x^2 + 1}}$

Ex-1 Let $x_0 = 0$

$$x_1 = \phi(x_0) = 1$$

$$\frac{x_1 - x_0}{x_2} = \phi(x_1) = \frac{1}{\sqrt{2}} = 0.7071$$

$$\frac{x_2 - x_1}{x_3} = \phi(x_2) = \frac{1}{\sqrt{0.7071}} = 0.76537$$

$$\underline{\text{Step-4}} \quad x_4 = \phi(x_3) = \frac{1}{\sqrt{1.76537}}$$

$$x_4 = 0.75263$$

$$\underline{\text{Step-5}} \quad x_5 = \phi(x_4) = \frac{1}{\sqrt{1.75263}}$$

$$x_5 = 0.75536$$

$$\underline{\text{Step-6}} \quad x_6 = \phi(x_5) = \frac{1}{\sqrt{1.75536}}$$

$$= 0.75477$$

$$\underline{\text{Step-7}} \quad x_7 = \phi(x_6) = \frac{1}{\sqrt{1.75477}}$$

$$x_7 = 0.75490$$

$$\underline{\text{Step-8}} \quad x_8 = \phi(x_7) = \frac{1}{\sqrt{1.75490}}$$

$$= 0.75487$$

$$\underline{\text{Step-9}} \quad x_9 = \phi(x_8) = \frac{1}{\sqrt{1.75487}}$$

$$= 0.75488$$

Upto 4 decimal places root is 0.7548

⑦

Descartes's Rule of Sign

Step 1: List out the signs of each variable below the expression.

Step 2: Note no. of changes in signs.

Step 3: $f(x)$ - sign changes = p
 $f(-x)$ - sign changes = q

Calculate using formula - no. of complex roots
 $= n - (p + q)$

n = highest power.

Q] $f(x) = x^5 - x^4 + 3x^3 + 9x^2 - x + 5$

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$$\begin{array}{cccccc} + & - & + & + & - & + \\ \downarrow & \uparrow & \downarrow & \uparrow & \downarrow & \uparrow \\ 1 & 2 & 3 & 4 & 5 & 6 \end{array}$$

$p = 4$

$f(-x) = -x^5 - x^4 - 3x^3 + 9x^2 + x + 5$

$$\begin{array}{cccccc} - & - & - & + & + & + \\ & & & \downarrow & \uparrow & \\ & & & 1 & 2 & \end{array}$$

$q = 1$

$$\begin{aligned} \text{No. of complex roots} &= n - (p + q) \\ &= 5 - (4 + 1) \\ &= 0 \end{aligned}$$

⑧ Budan's Theorem

Let $f(x) = 0$ be the given eqⁿ.

step 1: Find $f'(x)$ $f''(x)$.

(Hint: Find derivatives upto the highest power given in the eqⁿ).

step 2: Make tabular form.

x	$f(x)$	$f'(x)$	...	$f''(x)$	$V(x)$
$\downarrow$					$\downarrow$
intervals given	note the sign changes.				no. of sign changes

step 3: $V(\alpha) - V(\beta) = 0$ No real roots

$= 1$ Exactly 1 root

$= 2$ 2 or 0 real roots

$= 3$ 3 or 1 real roots

$= 4$ 4, 2, 0 real roots

$= 5$ 5, 3, 1 real roots

Q]

$$x^5 - x^4 + 3x^3 + 9x^2 - x + 5 = 0$$

$$[-3, -2], [-2, -1], [-1, 0], [0, 1]$$

$$f(x) = x^5 - x^4 + 3x^3 + 9x^2 - x + 5$$

$$f'(x) = 5x^4 - 4x^3 + 9x^2 + 18x - 1$$

$$f''(x) = 20x^3 - 12x^2 + 18x + 18$$

$$f'''(x) = 60x^2 - 24x + 18$$

$$f^{IV}(x) = 120x - 24$$

$$f^V(x) = 120$$

Table:-

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{IV}(x)$	$f^V(x)$	$v(x)$
-3	-316	+539	-684	+810	-744	240	5
-2	-29	+111	-226	+386	-504	+240	5
-1	+10	-1	-32	+122	-264	+240	4
0	+5	-1	+18	+18	-24	+240	4
1	+16	+27	+44	+74	+216	+240	0

$$- \quad v(-3) - v(-2) = 5 - 5 = 0$$

$$\Rightarrow \text{No real roots.}$$

$$- \quad v(-2) - v(-1) = 5 - 4 = 1$$

$$\Rightarrow 1 \text{ real root}$$

$$- \quad v(-1) - v(0) = 4 - 4 = 0$$

$$\Rightarrow \text{No real roots}$$

$$- \quad v(0) - v(1) = 4 - 0 = 4$$

$$\Rightarrow 4, 2 \text{ or } 0 \text{ real roots.}$$

⑨ Baristow's Method

step 1: Prepare division table

step 2: Find Δp_0 & Δq_0 as:-

$$\Delta p_0 = \frac{b_3 c_2 - b_4 c_1}{c_2^2 - c_1(c_3 - b_3)}$$

$$\Delta q_0 = \frac{b_4 c_2 - b_3(c_3 - b_3)}{c_2^2 - c_1(c_3 - b_3)}$$

step 3: Find new p & q

$$p = p_0 + \Delta p_0$$

$$q = q_0 + \Delta q_0$$

[Note: Take $-p$, $-q$ for the division table].

Q] $x^4 + x^3 + 2x^2 + x + 1 = 0$. $p_0 = q_0 = 0.5$
upto 1 iter.

Division Table

	1	1	2	1	1
$f(p)(-0.5)$	0	-0.5	-0.25	-0.6250	-0.0625
$f(q)(-0.5)$	0	0	-0.5	-0.2500	-0.6250
	1	0.5	1.25	0.1250	0.3125
	(b_0)	(b_1)	(b_2)	(b_3)	(b_4)
$f(p)(-0.5)$	0	-0.5	0	-0.3750	0
$q(-0.5)$	0	0	-0.5	0	
	1	0	0.7500	-0.2500	
	(c_0)	(c_1)	(c_2)	(c_3)	

$$\Delta p_0 = \frac{b_2 c_2 - b_4 c_1}{c_2^2 - b_3(c_3 - c_2)}$$

$$\Delta p_0 = \frac{b_3 c_2 - b_4 c_1}{c_2^2 - c_1(c_3 - b_3)}$$

$$\Delta p_0 = 0.1666.$$

$$\Delta q_0 = \frac{b_4 c_2 - b_3(c_3 - b_3)}{c_2^2 - c_1(c_3 - b_3)}$$

$$= 0.4997$$

$$P_1 = P_0 + \Delta P_0 = 0.5 + 0.1666 = 0.6666$$

$$q_1 = q_0 + \Delta q_0 = 0.5 + 0.4997 = 0.9997$$

⇒ LPn Baird Stow's METHOD (max do till $\text{Iter} - 2$)

eg. Find the quadratic factor $x^2 + Px + Q$ of the Polynomial $x^4 - 3x^3 + 20x^2 + 44x + 54$ using LPn Baird Stow's method. Take initial values of P & Q as 2. Consider 4 decimal in calculation.

Ans $f(x) = x^4 - 3x^3 + 20x^2 + 44x + 54$; $P_0 = Q_0 = 2$.

Always take initial P & Q as ± 1 or ± 2 .

Division Table.

	$(a_0) 1$	$(a_1) -3$	$(a_2) 20$	$(a_3) 44$	$(a_4) 54$	
$(P) -2$	0	-2	10	-5.6	4	
$(Q) -2$	0	0	-2	10	-5.6	
$(P) -2$	0	-2	14	-80		
$(Q) -2$	0	0	-2	+14		
	$(b_0) 1$	$(b_1) -5$	$(b_2) 28$	$(b_3) -2$	$(b_4) 2$	
	$(c_0) 1$	$(c_1) -7$	$(c_2) 40$	$(c_3) -68$		

from table

$$a_0 = 1 \quad a_1 = -3 \quad a_2 = 20 \quad a_3 = 44 \quad a_4 = 54$$

$$b_0 = 1 \quad b_1 = -5 \quad b_2 = 28 \quad b_3 = -2 \quad b_4 = 2$$

$$c_0 = 1 \quad c_1 = -7 \quad c_2 = 40 \quad c_3 = -68$$

now,

$$\Delta P_0 = \frac{b_3 C_2 - b_4 C_1}{C_2^2 - C_1(C_3 - b_2)}$$

$$= \frac{(-2)(40) - (2)(1)}{(40)^2 - (-7)(-68 - (-2))}$$

$$= \frac{-82}{1138} = -0.0580 \Rightarrow$$

$$\Delta Q_0 = \frac{b_4 C_2 - b_3(C_3 - b_2)}{C_2^2 - C_1(C_3 - b_2)}$$

$$= \frac{(2)(40) - (-2)(-66)}{1138}$$

$$= \frac{-52}{1138} = -0.04569$$

$$P_1 = P_0 + \Delta P_0 = 1.9420$$

$$Q_1 = Q_0 + \Delta Q_0 = 1.95434$$

$\therefore$ now new approximation are 1.9420, 1.95434

1st: $P \times b_0$
 $P \times b_1$

$P \times b_0 \dots$

$Q \times b_0 \dots$

Q1 (5)

- (1) State condⁿ of convergence in iteration method. |
- (2) State 2 differences b/w false position and N-R Method. |
- (3) Define accuracy & precision. |

Q2 (4) What is the rate of convergence of Bisection & N-R Method. |

- (5) Can you apply false position method to obtain root of eqⁿ: $xe^x - 2 = 0$ in $[0, 0.5]$. Justify your answer. |

Q2

- (1) Using Budan's theorem find the no. of roots of $x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1 = 0$ in the intervals $[-2, -1]$, $[-1, 0]$, $[0, 1]$, $[1, 2]$
- (2) Describe the types of errors occurring in numerical calculations. (2)

Q3

- (1) The -ve root of the smallest magnitude of eqⁿ $3x^3 + 10x^2 + 10x + 7 = 0$ is to be obtained.
- (5)
 - (i) Find an interval of unit length which contains this root
 - (ii) Perform 2 iterations of bisection method.
 - (iii) Taking the endpts of last interval as initial approx. perform 3 itr of Secant &

regula-falsi method.

(2) The smallest true root of eqⁿ: $x^4 - 3x^2 + x - 10 = 0$ is to be obtained.

(5) (i) ~~find~~ same as (i) } of (1).
(ii) same as (ii)

(iii) Taking midpt of last interval as initial approximation perform 3 iters of N-R Method.

x_0 = midpt of Bisection.

(3) Perform 1 iter Bairstow method to extract a quadratic factor $x^2 + px + q$ from

(5) polynomial $x^4 + x^3 + 2x^2 + x + 1 = 0$. Use initial approximations p_0 & $q_0 = 0.5$